MiSo: Mixture-of-Submanifolds Nonnegative Matrix Factorization for Interpretable Factor-Usage Motifs

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Abstract

Nonnegative matrix factorization is widely used to represent a nonnegative data matrix through low-rank nonnegative factors and loadings. Many variants encourage individual observations to use only a small number of factors, but they usually do not model recurring patterns of factor usage as inferential objects. We introduce MiSo (Mixture of Submanifolds), a probabilistic model for count data that learns recurring factor-usage motifs. Each observation is assigned softly to a latent motif, and each motif defines a low-dimensional submanifold spanned by a small subset of shared factor rows. The factor rows themselves may be dense. Each motif dimension has a posterior distribution over the shared factor rows, giving both clustering by factor-usage motif and posterior uncertainty over the factors defining each motif. The method builds on a Poisson SuSiE model for one count vector and a matrix-factorization extension of Poisson SuSiE. We describe the model, a variational empirical-Bayes fitting procedure, and the simulation studies needed to evaluate factor recovery, motif recovery, overfitting behavior, and posterior uncertainty.

1 Introduction

Nonnegative matrix factorization (NMF) seeks a low-rank representation of a nonnegative matrix $Y \in \mathbb{N}^{N\times M}$ by writing

$$Y \approx LF,$$

where $L \in \mathbb{R}_+^{N\times K}$ contains nonnegative loadings and $F \in \mathbb{R}_+^{K\times M}$ contains nonnegative factors. In many scientific applications the rows of L are expected to be sparse: each observation should be explained by only a few latent factors. Sparse NMF methods address this by regularizing or constraining the loading matrix. However, sparsity itself is not the whole inferential target. Often we want to know whether many observations share the same kind of sparsity pattern, which factors define that pattern, and whether the evidence distinguishes one possible factor combination from another.

This paper develops MiSo, a mixture-of-submanifolds model for nonnegative count matrix factorization. The main idea is to treat recurring factor-usage patterns as latent motifs. Each motif represents a small collection of factor rows that tend to appear together in the loadings. Observations are softly assigned to motifs, and each motif has posterior uncertainty over which factors define its dimensions. Thus the model is designed to answer questions such as: Which factor-usage motifs are present? Which observations use each motif? Which factors define a motif? Are two nearly identical or correlated factors interchangeable in the posterior?

The statistical challenge is that sparse NMF has several sources of non-identifiability. Factor rows can be permuted, highly correlated factors can compete to explain the same counts, and an overfitted factorization can produce duplicate or redundant factors. A hard sparse loading estimate can hide this ambiguity by choosing one arbitrary support. We instead represent uncertainty with posterior probabilities over factor identities, following the spirit of SuSiE [Wang et al., 2020]. The resulting posterior summaries are analogous to posterior inclusion probabilities, but they operate at the motif level rather than only at the observation level.

The paper has three methodological layers. Section 2 introduces Poisson SuSiE for one count vector and MF-Poisson-SuSiE for a count matrix. Section 3 introduces MiSo, which pools factor-usage patterns across observations through latent motifs. Section 4 is reserved for identifiability theory. Section 5 describes the experiments needed.

2 Poisson SuSiE and MF-Poisson-SuSiE

2.1 Single-vector Poisson SuSiE

We begin with one count vector $y \in \mathbb{N}^M$ and a known nonnegative dictionary $F \in \mathbb{R}_+^{K\times M}$. Throughout this paper we use row-normalized factors,

$$F_{km} \geq 0, \qquad \sum_{m=1}^{M} F_{km} = 1, \qquad k = 1, \ldots, K.$$

Poisson SuSiE represents the Poisson mean as a sparse sum of D loading dimensions:

$$y_m \mid \lambda, \gamma, F \sim \mathrm{Poi}(\mu_m), \qquad \mu_m = \sum_{d=1}^{D} \lambda_d F_{\gamma_d m}, \qquad m = 1, \ldots, M.$$

For each dimension d,

$$\gamma_d \sim \mathrm{Cat}(1/K, \ldots, 1/K), \qquad \lambda_d \mid \gamma_d \sim \mathrm{Gam}(\alpha_d^0, \beta_d^0),$$

where the Gamma distribution is parameterized by shape and rate. The latent index γ_d chooses one factor row, and λ_d is the nonnegative loading size for dimension d.

The variational approximation factorizes over dimensions:

$$q(\lambda, \gamma) = \prod_{d=1}^{D} q(\gamma_d) q(\lambda_d \mid \gamma_d),$$

with

$$q(\gamma_d = k) = \bar{\gamma}_{dk}, \qquad q(\lambda_d \mid \gamma_d = k) = \mathrm{Gam}(\alpha_{dk}, \beta_{dk}).$$

We also introduce allocation variables ξ_{dm} satisfying $\xi_{dm} \geq 0$ and $\sum_{d=1}^{D} \xi_{dm} = 1$ for each coordinate m. They allocate the observed count y_m across the D dimensions. Jensen's inequality gives the lower bound contribution

$$y_m \log\left(\sum_{d=1}^{D} \lambda_d F_{\gamma_d m}\right) \geq \sum_{d=1}^{D} \xi_{dm} y_m \log\left(\frac{\lambda_d F_{\gamma_d m}}{\xi_{dm}}\right).$$

The resulting coordinate updates have a simple form. The allocation update is

$$\xi_{dm} \propto \exp\left[\sum_{k=1}^{K} \bar{\gamma}_{dk}\{\psi(\alpha_{dk}) - \log \beta_{dk}\} + \sum_{k=1}^{K} \bar{\gamma}_{dk} \log F_{km}\right],$$

normalized over $d = 1, \ldots, D$. The Gamma variational parameters are

$$\alpha_{dk} \leftarrow \alpha_d^0 + \sum_{m=1}^{M} \xi_{dm} y_m, \qquad \beta_{dk} \leftarrow \beta_d^0 + \sum_{m=1}^{M} F_{km}.$$

The posterior probability that dimension d uses factor k is updated by a softmax over the terms in the lower bound depending on $\gamma_d = k$:

$$\bar{\gamma}_{dk} \propto \exp\left[\sum_{m=1}^{M} \xi_{dm} y_m \log F_{km} - \left(\alpha_d^0 + \sum_{m=1}^{M} \xi_{dm} y_m\right) \log\left(\beta_d^0 + \sum_{m=1}^{M} F_{km}\right)\right].$$

When the rows of F are normalized, the rate term is constant across k, but we keep it visible because it clarifies the role of the Gamma prior and the scale convention.

2.2 Empirical-Bayes prior update

Poisson SuSiE can use empirical Bayes to update the prior mean of each loading dimension. For dimension d, define the posterior mean and posterior mean log-size

$$\sum_{k=1}^{K} \bar{\gamma}_{dk} \frac{\alpha_{dk}}{\beta_{dk}}, \qquad \sum_{k=1}^{K} \bar{\gamma}_{dk} \{\psi(\alpha_{dk}) - \log \beta_{dk}\}.$$

Maximizing the expected Gamma prior gives the rate update

$$\beta_d^0 = \frac{\alpha_d^0}{\sum_{k=1}^{K} \bar{\gamma}_{dk} \alpha_{dk} / \beta_{dk}},$$

and α_d^0 solves

$$\psi(\alpha_d^0) = \sum_{k=1}^{K} \bar{\gamma}_{dk} \{\psi(\alpha_{dk}) - \log \beta_{dk}\} + \log \beta_d^0.$$

This empirical-Bayes step is important when D is overspecified: dimensions with little evidence can have prior mean α_d^0/β_d^0 close to zero, which is an automatic relevance determination (ARD) mechanism for selecting the effective number of dimensions.

2.3 MF-Poisson-SuSiE

MF-Poisson-SuSiE applies the same sparse Poisson representation to each row of a count matrix while learning the shared factor matrix F. For observation $i = 1, \ldots, N$,

$$Y_{im} \mid \lambda_i, \gamma_i, F \sim \text{Poi}\left(\sum_{d=1}^{D} \lambda_{id} F_{\gamma_{id} m}\right), \qquad m = 1, \ldots, M.$$

Each observation has its own sparse support indicators γ_{id} and loading sizes λ_{id}. The variational posterior uses

$$q(\gamma_{id} = k) = \bar{\gamma}_{idk}, \qquad q(\lambda_{id} \mid \gamma_{id} = k) = \text{Gam}(\alpha_{idk}, \beta_{idk}).$$

For fixed F, the updates are the row-wise analogues of the single-vector updates. When F is unknown, the factor update uses expected allocated counts:

$$A_{km} = \sum_{i=1}^{N} \sum_{d=1}^{D} \bar{\gamma}_{idk} \xi_{idm} Y_{im}.$$

With row-normalized factors,

$$F_{km} \leftarrow \frac{A_{km} + \epsilon}{\sum_{m'=1}^{M}(A_{km'} + \epsilon)}.$$

In practice the implementation initializes F with Poisson NMF, initializes the sparse posterior from the corresponding loading estimates, and damps the updates of $\bar{\gamma}$ and F. This produces a useful intermediate model: it learns the factor dictionary and observation-level sparse loading uncertainty, but it does not yet pool recurring support patterns across observations. That pooling is the purpose of MiSo.

3 MiSo

3.1 Generative model

MiSo models repeated factor-usage motifs. The observed data are counts $Y \in \mathbb{N}^{N\times M}$. The potentially dense factor matrix $F \in \mathbb{R}_+^{K\times M}$ is row-normalized as above. Each motif identifies a low-dimensional submanifold spanned by a subset of these shared factor rows; the model does not impose sparsity on F. There are S latent motifs. We initially fit each motif with a nominal dimension D, but the effective dimension is allowed to be smaller and can vary by motif.

For motif s and dimension d, the factor identity is random:

$$\gamma_{sd} \sim \mathrm{Cat}(\rho_{sd1}, \ldots, \rho_{sdK}).$$

For observation i,

$$Z_i \sim \mathrm{Cat}(\pi_1, \ldots, \pi_S).$$

Conditional on $Z_i = s$, the observation has one nonnegative loading size for each dimension:

$$\lambda_{id} \mid Z_i = s \sim \mathrm{Gam}(\alpha^0_{sd}, \beta^0_{sd}), \qquad d = 1, \ldots, D.$$

The observed counts are

$$Y_{im} \mid Z_i = s, \lambda_i, \gamma, F \sim \mathrm{Poi}(\mu_{im}), \qquad \mu_{im} = \sum_{d=1}^{D} \lambda_{id} F_{\gamma_{sd} m}, \qquad m = 1, \ldots, M.$$

Equivalently, if L_i denotes the loading vector for observation i, then

$$L_i = \sum_{d=1}^{D} \lambda_{id} e_{\gamma_{Z_i d}},$$

where e_k is the kth coordinate vector in $\mathbb{R}^K$. Thus all observations assigned to motif s share the same motif-level factor identities $\gamma_{s1}, \ldots, \gamma_{sD}$, but each observation has its own loading sizes.

3.2 Variational family

The variational approximation is

$$q(Z, \gamma, \lambda) = \prod_{i=1}^{N} q(Z_i) \prod_{s=1}^{S}\prod_{d=1}^{D} q(\gamma_{sd}) \prod_{i=1}^{N}\prod_{s=1}^{S}\prod_{d=1}^{D} q(\lambda_{id} \mid Z_i = s).$$

We write

$$q(Z_i = s) = \omega_{is}, \qquad q(\gamma_{sd} = k) = \bar{\gamma}_{sdk},$$

and

$$q(\lambda_{id} \mid Z_i = s) = \mathrm{Gam}(\alpha_{isd}, \beta_{sd}).$$

The rate parameter β_{sd} does not depend on i because the Poisson factor rows are shared and row-normalized; the shape parameter α_{isd} depends on the counts allocated from observation i.

For each candidate motif s, we introduce allocation variables ξ_{isdm} satisfying

$$\xi_{isdm} \geq 0, \qquad \sum_{d=1}^{D} \xi_{isdm} = 1.$$

They allocate Y_{im} across the dimensions of motif s. The implementation computes ξ_{isdm} in feature blocks rather than storing the full $N \times S \times D \times M$ tensor.

3.3 Coordinate updates

The allocation update is

$$\xi_{isdm} \propto \exp\left[\psi(\alpha_{isd}) - \log \beta_{sd} + \sum_{k=1}^{K} \bar{\gamma}_{sdk} \log F_{km}\right],$$

normalized over $d = 1, \ldots, D$. The Gamma posterior update is

$$\alpha_{isd} \leftarrow \alpha_{sd}^{0} + \sum_{m=1}^{M} \xi_{isdm} Y_{im},$$

and

$$\beta_{sd} \leftarrow \beta_{sd}^{0} + \sum_{m=1}^{M} \sum_{k=1}^{K} \bar{\gamma}_{sdk} F_{km}.$$

The second term is equal to one when each row of F is normalized, but we write the full expression to make the model dependence explicit.

For motif dimension (s, d), the posterior probability of factor k is updated by

$$\bar{\gamma}_{sdk} \propto \rho_{sdk} \exp\left[\sum_{i=1}^{N} \omega_{is} \left\{\sum_{m=1}^{M} \xi_{isdm} Y_{im} \log F_{km} - \frac{\alpha_{isd}}{\beta_{sd}} \sum_{m=1}^{M} F_{km}\right\}\right].$$

This is the central posterior uncertainty object in MiSo. A concentrated vector $\bar{\gamma}_{sd1}, \ldots, \bar{\gamma}_{sdK}$ means that motif dimension (s, d) has a well-identified factor. A diffuse vector means that several factor rows are plausible explanations for the same motif dimension.

The motif responsibility for observation i is updated from the component lower bound for that observation under each motif:

$$\omega_{is} \propto \pi_s \exp\{\mathcal{L}_{is}\},$$

where

$$\begin{aligned}\mathcal{L}_{is} = &\sum_{d=1}^{D} \sum_{m=1}^{M} \xi_{isdm} Y_{im} \left\{\psi(\alpha_{isd}) - \log \beta_{sd} + \sum_{k=1}^{K} \bar{\gamma}_{sdk} \log F_{km} - \log \xi_{isdm}\right\} \\ &- \sum_{d=1}^{D} \sum_{m=1}^{M} \frac{\alpha_{isd}}{\beta_{sd}} \sum_{k=1}^{K} \bar{\gamma}_{sdk} F_{km} - \sum_{d=1}^{D} \mathrm{KL}\{\mathrm{Gam}(\alpha_{isd}, \beta_{sd}) \,\|\, \mathrm{Gam}(\alpha_{sd}^{0}, \beta_{sd}^{0})\}.\end{aligned}$$

The mixture weights are updated by

$$\pi_s \leftarrow \frac{1}{N}\sum_{i=1}^{N}\omega_{is}.$$

3.4 Learning the factors

When F is unknown, MiSo updates F using the expected allocated counts

$$A_{km} = \sum_{i=1}^{N}\sum_{s=1}^{S}\sum_{d=1}^{D}\omega_{is}\bar{\gamma}_{sdk}\xi_{isdm}Y_{im}.$$

The row-normalized update is

$$F_{km} \leftarrow \frac{A_{km}+\epsilon}{\sum_{m'=1}^{M}(A_{km'}+\epsilon)}.$$

In the current implementation, F is initialized by MF-Poisson-SuSiE with learned F, and the subsequent MiSo updates alternate between motif responsibilities, motif posteriors, Gamma loading posteriors, empirical-Bayes prior updates, and optionally factor updates. This initialization is important because the full problem is highly nonconvex. We next specify the initialization used to convert observation-level preliminary loadings into motif-level factor directions.

3.5 Initialization

MiSo is initialized from an MF-Poisson-SuSiE fit with learned factors. Let $\bar{\gamma}^{\mathrm{MF}}_{idk}$, $\alpha^{\mathrm{MF}}_{idk}$, and $\beta^{\mathrm{MF}}_{idk}$ denote its variational parameters. The preliminary expected loading of factor k in observation i is

$$\widetilde{L}_{ik} = \sum_{d=1}^{D}\bar{\gamma}^{\mathrm{MF}}_{idk}\frac{\alpha^{\mathrm{MF}}_{idk}}{\beta^{\mathrm{MF}}_{idk}}.$$

We remove observation-level scale for the purpose of constructing initial motifs,

$$x_{ik} = \frac{\widetilde{L}_{ik}}{\sum_{h=1}^{K}\widetilde{L}_{ih}}, \qquad x_i \in \Delta^{K-1},$$

and obtain S preliminary centers by solving

$$(\widehat{z},\widehat{c}) = \arg\min_{z_1,\ldots,z_N,\,c_1,\ldots,c_S}\sum_{i=1}^{N}\|x_i - c_{z_i}\|_2^2, \qquad c_s \in \Delta^{K-1}.$$

Thus the preliminary loadings are observation-specific, whereas the factor directions used to initialize MiSo are selected from the center of each candidate motif.

For motif s, let

$$r_s(1),\ldots,r_s(K)$$

order the factors so that

$$c_{s,r_s(1)} \geq c_{s,r_s(2)} \geq \cdots \geq c_{s,r_s(K)}.$$

Assuming $D \leq K$, MiSo uses a distinct top-D initialization,

$$\kappa^{(0)}_{sd} = r_s(d), \qquad d = 1,\ldots,D, \tag{1}$$

and initializes the motif-level factor posterior by

$$\bar{\gamma}_{sdk}^{(0)} = (1 - \varepsilon_\gamma)\mathbf{1}\{k = \kappa_{sd}^{(0)}\} + \frac{\varepsilon_\gamma}{K}, \qquad k = 1, \dots, K, \tag{2}$$

for a small $\varepsilon_\gamma > 0$. The implementation uses $\varepsilon_\gamma = 0.05$ by default. The ordering in (1) is only a deterministic convention; the motif dimensions remain permutation invariant.

Because r_s is a permutation and $D \leq K$, the indices $\kappa_{s1}^{(0)}, \dots, \kappa_{sD}^{(0)}$ are pairwise distinct. Thus every motif begins with D different candidate factor directions. If D is over-specified, unnecessary directions are removed later through the empirical-Bayes dimension diagnostic.

The initial mixture weights and responsibilities are set to

$$\pi_s^{(0)} = \frac{1}{S}, \qquad \omega_{is}^{(0)} = \frac{1}{S}.$$

The preliminary partition is consequently used to construct motif centers, but is not imposed as a hard initial assignment of observations to motifs. The Gamma variational and prior parameters are initialized symmetrically across motifs and are then updated using the raw count matrix.

3.6 Empirical-Bayes dimension selection

Each fitted motif starts with the same nominal dimension D, but the effective dimension of motif s can be smaller. MiSo uses the empirical-Bayes Gamma prior parameters to diagnose dimension activity. For each motif s and dimension d, the prior mean

$$\frac{\alpha_{sd}^0}{\beta_{sd}^0}$$

estimates the loading size expected for that dimension. If this value is close to zero relative to the other dimensions in motif s, then dimension d is effectively inactive. This gives an ARD-style selected dimension

$$\widehat{D}_s = \#\left\{ d : \frac{\alpha_{sd}^0/\beta_{sd}^0}{\sum_{d'=1}^{D} \alpha_{sd'}^0/\beta_{sd'}^0} \geq \tau \right\},$$

for a threshold τ.

This ARD diagnostic should be interpreted separately from the entropy of $\bar{\gamma}_{sd1}, \dots, \bar{\gamma}_{sdK}$. The ARD prior mean asks whether a dimension carries loading mass. The entropy of $\bar{\gamma}_{sd}$ asks whether the factor identity of that dimension is uncertain. Both are useful: an inactive dimension should have small loading mass, while an active but ambiguous dimension may have large loading mass and diffuse posterior factor identity.

4 Identifiability Theory

MiSo places its geometric structure on the latent Poisson intensity, not on the observed count vector. This distinction lets us retain the language of submanifolds while using cones when a closed support or an extreme ray is the mathematically relevant object. For a realized motif s, write

$$H_s = \sum_{d=1}^{D_s} a_{sd} \delta_{\theta_{sd}}, \qquad \theta_{sd} = \frac{F_{\gamma_{sd}}}{\beta_{sd}}, \qquad a_{sd}, \beta_{sd} > 0, \tag{3}$$

Algorithm 1 MiSo variational empirical-Bayes fitting

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Fit MF-Poisson-SuSiE to initialize F and preliminary loadings L̃.
Cluster normalized preliminary loadings into S centers c_s.
Initialize γ̄_sdk from the distinct top D factors of each center using (2).
Initialize ω_is = 1/S and π_s = 1/S.
Initialize α_isd, β_sd, and (α⁰_sd, β⁰_sd).
for t = 1, ..., T do
    Update ξ_isdm in feature blocks.
    Update α_isd and β_sd.
    Update ω_is and π_s.
    Update (α⁰_sd, β⁰_sd) by empirical Bayes.
    Update γ̄_sdk for each motif dimension.
    if factor learning is enabled then
        Update F using expected allocated counts and row-normalize.
    end if
    Monitor the variational objective or a scheduled approximation to it.
end for
Report motif responsibilities ω, motif posteriors γ̄, selected dimensions D̂_s, and fitted factors F.
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where a_{sd} and β_{sd} are the Gamma shape and rate. Coincident atoms are combined by adding their shapes, giving the canonical representation

$$H = \sum_{j=1}^{J} a_j \delta_{\theta_j}, \qquad a_j > 0, \qquad \theta_1, \ldots, \theta_J \text{ pairwise distinct.} \tag{4}$$

The corresponding latent submanifold and its closure cone are

$$\mathcal{M}(H) = \left\{ \sum_{j=1}^{J} x_j \theta_j : x_j > 0 \right\}, \tag{5}$$

$$\mathcal{C}(H) = \overline{\mathcal{M}(H)} = \operatorname{cone}\{\theta_1, \ldots, \theta_J\}. \tag{6}$$

Thus $\mathcal{M}(H)$ is the relative interior of $\mathcal{C}(H)$ and is a smooth submanifold of its linear span. Its intrinsic dimension is $\dim \mathcal{M}(H) = \operatorname{rank}(\theta_1, \ldots, \theta_J)$, which can be smaller than the nominal number of dimensions D_s.

To define the distribution encoded by H, independently draw

$$\xi_j \sim \operatorname{Gam}(a_j, 1), \qquad j = 1, \ldots, J,$$

and set

$$\mu_H = \sum_{j=1}^{J} \xi_j \theta_j. \tag{7}$$

We denote the law of μ_H by $\operatorname{Gam}(H)$. This agrees with the original shape–rate parameterization because, if $\lambda_j \sim \operatorname{Gam}(a_j, \beta_j)$, then $\beta_j \lambda_j \sim \operatorname{Gam}(a_j, 1)$ and $\lambda_j F_{\gamma_j} = (\beta_j \lambda_j)(F_{\gamma_j}/\beta_j)$. Conditional on μ_H, the coordinates of Y are independent with

$$Y_m \mid \mu_H \sim \operatorname{Poi}((\mu_H)_m), \qquad m = 1, \ldots, M.$$

Although $\mu_H \in \mathcal{M}(H)$ almost surely, the topological support of its distribution is the closure cone $\mathcal{C}(H)$.

The population mixture of latent submanifolds is represented by

$$G = \sum_{s=1}^{S} \pi_s \delta_{H_s}, \qquad \pi_s > 0, \qquad \sum_{s=1}^{S} \pi_s = 1, \tag{8}$$

and its observed count law is

$$\mathrm{MiSo}(G) = \sum_{s=1}^{S} \pi_s \mathrm{Poi}\{\mathrm{Gam}(H_s)\}. \tag{9}$$

The motif measure H_s need not be a probability measure: its atom masses are Gamma shapes, whereas π_s is the mixture probability. For $t \in \mathbb{R}_+^M$, define

$$\Phi_H(t) := \mathbb{E}\exp\{-\langle t, \mu_H\rangle\} = \prod_{j=1}^{J} (1 + \langle \theta_j, t\rangle)^{-a_j}. \tag{10}$$

The probability-generating function of the noisy counts is

$$\mathbb{E}_H\left[\prod_{m=1}^{M} z_m^{Y_m}\right] = \Phi_H(\mathbf{1} - z), \qquad z \in [0,1]^M. \tag{11}$$

This identity is the bridge from the observed Poisson law to the geometry of the latent submanifolds.

Theorem 1 (Identification of one latent submanifold)**.** *Let H and H' be finite positive atomic measures on $\mathbb{R}_+^M \setminus \{0\}$. Then*

$$\mathrm{Poi}\{\mathrm{Gam}(H)\} \overset{d}{=} \mathrm{Poi}\{\mathrm{Gam}(H')\} \qquad \Longrightarrow \qquad H = H'.$$

Consequently, the latent submanifold $\mathcal{M}(H)$, its closure cone $\mathcal{C}(H)$, and its intrinsic dimension are identifiable from the population distribution of the Poisson observations.

Proof. Equality of the count laws gives equality of their probability-generating functions. By (11), $\Phi_H(t) = \Phi_{H'}(t)$ for $t \in (0,1)^M$. Hence their negative log-Laplace transforms agree:

$$\sum_j a_j \log(1 + \langle \theta_j, t\rangle) = \sum_\ell a'_\ell \log(1 + \langle \theta'_\ell, t\rangle).$$

Differentiating on this open set gives

$$\sum_j \frac{a_j \theta_j}{1 + \langle \theta_j, t\rangle} = \sum_\ell \frac{a'_\ell \theta'_\ell}{1 + \langle \theta'_\ell, t\rangle}. \tag{12}$$

Choose $v \in \mathbb{R}_{++}^M$ such that distinct vectors among the finitely many θ_j and θ'_ℓ have distinct projections onto v. Such a v exists because the excluded choices lie in finitely many proper hyperplanes. Put $r_j = \langle \theta_j, v\rangle$ and $r'_\ell = \langle \theta'_\ell, v\rangle$, all of which are positive. On the real line $t = xv$, equation (12) becomes

$$\sum_j \frac{a_j \theta_j / r_j}{x + 1/r_j} = \sum_\ell \frac{a'_\ell \theta'_\ell / r'_\ell}{x + 1/r'_\ell}$$

for x in a positive interval. Multiplication by all denominators gives an identity between vector-valued real polynomials. Therefore the two real partial-fraction decompositions have the same denominator roots and the same coefficient at each root. For a matched pair with $r_j = r'_\ell$, multiplying by $x + 1/r_j$ and taking the real limit $x \to -1/r_j$ gives

$$\frac{a_j \theta_j}{r_j} = \frac{a'_\ell \theta'_\ell}{r_j}.$$

Taking the inner product with v yields $a_j = a'_\ell$, after which $\theta_j = \theta'_\ell$. The generic choice of v makes the matching one-to-one, proving $H = H'$. $\square$

The measure representation absorbs the benign Gamma-splitting ambiguity: $a_1\delta_\theta + a_2\delta_\theta$ and $(a_1 + a_2)\delta_\theta$ are literally the same measure. Theorem 1 also makes explicit that the Poisson layer is lossless for population identification, because the count probability- generating function reveals the Laplace transform of the latent intensity distribution [Teicher, 1960].

For the mixture geometry, let

$$P_G = \sum_{s=1}^{S} \pi_s \mathrm{Gam}(H_s) \tag{13}$$

be the distribution of the latent intensity and define

$$\mathcal{U}(G) = \mathrm{supp}(P_G), \qquad \mathcal{C}(G) = \mathrm{cone}\{\mathcal{U}(G)\}.$$

The set $\mathcal{U}(G)$ retains the union of the closure cones of the latent submanifolds, whereas $\mathcal{C}(G)$ is their global conic hull.

Theorem 2 (Identification of support geometry and extreme rays)**.** *Let G and G' be finite probability measures of the form (8). If*

$$\mathrm{MiSo}(G) \overset{d}{=} \mathrm{MiSo}(G'),$$

then

$$P_G = P_{G'}, \qquad \mathcal{U}(G) = \mathcal{U}(G'), \qquad \mathcal{C}(G) = \mathcal{C}(G').$$

Consequently, the extreme rays of the global closure cone are identifiable despite the Poisson observation noise. If rays are normalized by their intersection with $\Delta^{M-1} = \{x \in \mathbb{R}^M_+ : \mathbf{1}^\top x = 1\}$, the corresponding extreme points are identifiable as well. Moreover, every global extreme ray contains an atom from at least one motif measure in the support of G and from at least one motif measure in the support of G'.

Proof. The count probability-generating function is the Laplace transform of P_G evaluated at $\mathbf{1} - z$. Equality of the MiSo laws therefore gives equality of the multivariate Laplace transforms of P_G and $P_{G'}$ on $(0, 1)^M$. Uniqueness of the Laplace transform on $\mathbb{R}^M_+$ gives $P_G = P_{G'}$.

For $H = \sum_j a_j \delta_{\theta_j}$, the independent Gamma coefficients in (7) have joint topological support $\mathbb{R}^J_+$. Their linear image consequently has support

$$\mathrm{supp}\{\mathrm{Gam}(H)\} = \left\{ \sum_j x_j \theta_j : x_j \geq 0 \right\} = \mathcal{C}(H).$$

Because the mixture is finite and every mixture weight is positive,

$$\mathcal{U}(G) = \bigcup_{H \in \mathrm{supp}(G)} \mathcal{C}(H). \tag{14}$$

The same identity holds for G'. Equality of P_G and $P_{G'}$ therefore identifies the union of closure cones and its global conic hull.

Extreme rays and normalized extreme points are intrinsic properties of this identified closed cone. Finally, $\mathcal{C}(G)$ is generated by the finite union of the atoms in the component measures. Every extreme ray of a finitely generated cone contains a vector from any finite generating set; otherwise a point on that ray would be a nontrivial conic combination of vectors off the ray. Applying this fact to the atoms under G and under G' proves the final claim. □

When each factor row is normalized by $\mathbf{1}^\top F_k = 1$, the scaled atom $\theta = F_k/b$ lies on the same ray as F_k, and simplex normalization maps it back to F_k. Theorem 2 therefore identifies the factor rows corresponding to global extreme rays. It does not, by itself, identify interior dictionary rows or assign the identified rays to individual motifs.

Remark 1 (Interior rays prevent unrestricted mixture identification)**.** Theorem 1 makes the single-submanifold kernel injective, but it does not imply that an unrestricted mixture G is identifiable. To see this with all Gamma shapes equal to one, choose linearly independent $\theta_A, \theta_B \in \mathbb{R}_+^M \setminus \{0\}$, fix $\rho \in (0,1)$, and introduce the interior direction

$$\theta_C = \rho\theta_A + (1-\rho)\theta_B.$$

Let

$$H_{AB} = \delta_{\theta_A} + \delta_{\theta_B}, \qquad H_{BC} = \delta_{\theta_B} + \delta_{\theta_C}, \qquad H_{AC} = \delta_{\theta_A} + \delta_{\theta_C}.$$

Writing $A(t) = 1 + \langle \theta_A, t\rangle$ and defining $B(t)$ and $C(t)$ analogously, we have $C = \rho A + (1-\rho)B$ and hence

$$\begin{aligned}\Phi_{H_{AB}}(t) &= \frac{1}{A(t)B(t)} \\ &= \frac{\rho}{B(t)C(t)} + \frac{1-\rho}{A(t)C(t)} \\ &= \rho\Phi_{H_{BC}}(t) + (1-\rho)\Phi_{H_{AC}}(t).\end{aligned} \tag{15}$$

Thus the distinct mixing measures

$$G = \delta_{H_{AB}}, \qquad G' = \rho\delta_{H_{BC}} + (1-\rho)\delta_{H_{AC}}$$

produce exactly the same noisy count law. Geometrically, one latent submanifold has been represented by a mixture whose two closure cones subdivide the original cone along an interior ray. This does not contradict Theorem 2: both representations have the same latent support and the same global extreme rays. Rather, it shows why an over-specified interior factor can prevent identification of the individual motifs even when the global geometry is identifiable.

To state the condition needed for full mixture identification, call a transform family $\{\Phi_H : H \in \mathcal{H}_0\}$ finitely linearly independent if, for every finite collection of distinct $H_1, \ldots, H_r \in \mathcal{H}_0$,

$$\sum_{j=1}^{r} c_j \Phi_{H_j}(t) = 0 \text{ for every } t \in (0,1)^M \quad \Longrightarrow \quad c_1 = \cdots = c_r = 0. \tag{16}$$

Theorem 3 (Finite-mixture identification)**.** *Let G and G' be finite probability measures on $\mathcal{H}_0$. If $\{\Phi_H : H \in \mathcal{H}_0\}$ is finitely linearly independent, then*

$$\mathrm{MiSo}(G) \overset{d}{=} \mathrm{MiSo}(G') \qquad \Longrightarrow \qquad G = G'.$$

Conversely, if the transform family is not finitely linearly independent, then the class of finite probability mixtures on $\mathcal{H}_0$ is not identifiable.

Proof. Take the union of the finite supports of G and G' and denote its distinct elements by $H_1, \dots, H_r$. Equality of the noisy count laws and (11) gives

$$\sum_{j=1}^{r} \{G(H_j) - G'(H_j)\} \Phi_{H_j}(t) = 0, \qquad t \in (0,1)^M.$$

Finite linear independence forces every coefficient to vanish, so $G = G'$. Conversely, consider a nonzero finite linear relation among distinct transforms. At $t = 0$, every $\Phi_H(0) = 1$, so the positive and negative coefficient sums are equal. Separating the two sides and normalizing by their common sum produces two different probability mixing measures with the same count probability-generating function, and hence the same MiSo law. $\square$

Theorem 3 is the standard finite-mixture criterion of Yakowitz and Spragins [1968], within the broader identifiability program of Teicher [1960, 1963]. It separates two questions that are sometimes conflated: injectivity of each component kernel, established by Theorem 1, and linear independence of the whole component family, which is required to identify G.

Corollary 4 (Common simplicial dictionary)**.** *Fix linearly independent vectors $\theta_1, \dots, \theta_K \in \mathbb{R}_+^M \setminus \{0\}$ with $K \le M$, write $\Theta = \{\theta_1, \dots, \theta_K\}$, and consider*

$$\mathcal{H}_\Theta = \left\{ H_\alpha = \sum_{k=1}^{K} \alpha_k \delta_{\theta_k} : \alpha \in \mathbb{R}_+^K \setminus \{0\} \right\}, \tag{17}$$

where a zero coordinate means that the corresponding atom is omitted. Then every finite MiSo mixing measure G on $\mathcal{H}_\Theta$ is identifiable.

Proof. For $H_\alpha \in \mathcal{H}_\Theta$,

$$\Phi_{H_\alpha}(t) = \prod_{k=1}^{K} (1 + \langle \theta_k, t \rangle)^{-\alpha_k}.$$

Because the θ_k are linearly independent, the map $t \mapsto (\langle \theta_1, t \rangle, \dots, \langle \theta_K, t \rangle)$ has full row rank and maps an open subset of $(0,1)^M$ onto an open subset of $\mathbb{R}^K$. Make the coordinate change

$$y_k = \log(1 + \langle \theta_k, t \rangle).$$

A finite linear relation among transforms associated with distinct $\alpha^{(j)}$ becomes

$$\sum_j c_j \exp\{-\langle \alpha^{(j)}, y \rangle\} = 0$$

on an open set. Choose a direction v for which the finitely many values $\langle \alpha^{(j)}, v \rangle$ are distinct and restrict the identity to a short real line segment parallel to v. Distinct univariate exponentials are linearly independent, so every $c_j = 0$. The result follows from Theorem 3. $\square$

Corollary 4 allows a motif to omit a global ray by setting its corresponding shape to zero. It requires the scaled direction θ_k to be shared across motifs, as occurs when selecting factor row F_k also selects a factor-specific rate β_k. The result also permits a fixed over-specified dictionary with $K' > K$ provided all K' scaled directions remain linearly independent, so necessarily $K' \le M$. The true model is then embedded by assigning zero shape to the extra directions. However, the corollary conditions on a fixed dictionary; it does not by itself establish joint identifiability when the dictionary is learned.

5 Experiments Needed

The empirical goal is not only to show good reconstruction error. The paper needs to show that MiSo recovers meaningful factor-usage motifs, clusters observations by motif, learns factors when F is unknown, and quantifies uncertainty in settings where hard sparse supports are unstable.

5.1 Baselines

The main baselines should be:

1. Poisson NMF followed by clustering of normalized loading scores;
2. MF-Poisson-SuSiE followed by clustering of posterior loading scores;
3. sparse NMF baselines with tuned sparsity penalties, if available;
4. ablations of MiSo with fixed versus learned F.

The first two baselines are already implemented in the current benchmark code.

5.2 Metrics

The simulation tables should include:

1. factor recovery, measured by cosine matching between fitted rows of F and true rows of F_0;
2. motif clustering accuracy, measured by matching $\arg\max_s \omega_{is}$ to true groups;
3. support recovery, measured by converting posterior motif scores to predicted active factors;
4. soft clustering diagnostics, based on group-level averages of ω_{is} rather than only hard labels;
5. posterior uncertainty summaries for $\bar{\gamma}_{sd}$, including entropy, top-two gaps, and targeted pairwise split metrics;
6. selected motif dimensions $\widehat{D}_s$ from the ARD prior means;
7. predictive fit, such as held-out Poisson log likelihood or deviance.

5.3 Core simulation settings

The current roadmap suggests the following simulation studies.

Correctly specified block motifs. Use correct K and nominal D, with groups occupying simple single-factor and multi-factor supports. This verifies that the method can recover factors, clusters, and supports in a clean setting. Current evidence shows feasibility, but a larger seed grid is needed before claiming robust superiority over baselines.

Tree/no-anchor motifs. Use motifs such as $(1, 2, 3)$, $(1, 2, 4)$, $(1, 5, 6)$, $(1, 5, 7)$, $(1, 2)$, and $(1, 5)$, so that no group is explained by a single anchor factor. This is the strongest current positive setting: in a small multi-seed pilot, MiSo has substantially better support recovery than the baseline pipelines.

Overspecified K. Generate data with K true factors but fit with extra candidate factor rows, including near-duplicates of true factors. The key diagnostic is whether $\bar{\gamma}_{sd}$ splits posterior mass across duplicate rows when they are statistically interchangeable. Global entropy is not sufficient; the paper should report pair-specific posterior split metrics for the duplicated factors.

Correlated factors. Generate correlated factor rows at controlled strengths. This tests whether MiSo can express targeted ambiguity between correlated factors. The current evidence suggests that raw entropy can be misleading because weak dimensions may become globally diffuse. The final paper version should sweep correlation strength and report targeted pair uncertainty, not only mean entropy.

Overspecified D. Fit each motif with a nominal dimension larger than the true motif dimension. The target behavior is that extra dimensions either have small ARD prior mean $\alpha^0_{sd}/\beta^0_{sd}$ or small posterior loading mass. The current default initializes the top D center factors without replacement. A full paper needs multi-seed threshold sweeps for $\widehat{D}_s$ and comparisons against the true motif dimensions after matching fitted motifs to groups.

5.4 Real data

The real-data analysis should demonstrate three advantages of the MiSo representation beyond reconstruction quality. First, it should identify recurring motif types: scientifically meaningful combinations of factors that are reused across observations. A small atlas of these motifs, together with their responsibilities, should provide a more concise summary than a typical admixture bar plot with many separately colored factors. Second, it should show that a motif represents a low-dimensional submanifold rather than merely a cluster. In the single-cell application, cells assigned to the same motif should populate its posterior dimension coordinates continuously, with related cell states, marker expression, and—where available—spatial position varying smoothly along a trajectory consistent with differentiation. Third, the analysis should recover structure among the factors themselves. Factor co-use across motifs induces a graph in which broadly reused factors may act as hubs and more specialized factors as leaves; these relationships should be checked against external biological interpretation.

These are goals for the final analysis rather than claims established by the current experiment. The identifiability theory remains to be completed, and the post-processing needed when K, S, and D are over-specified still requires further study. In particular, the current Wasserstein dendrogram and merging rule are exploratory and should not yet be presented as the final procedure.

A concise empirical demonstration. The main real-data result can be organized as one multi-panel figure. The first panel should show the conventional NMF loading bar plot and its loss of readability when K is moderately large. The second should replace it with a small motif atlas, displaying the factors and effective dimension used by each recurring motif together with its prevalence. The third should highlight one well-supported two- or three-dimensional motif and plot its cells in the corresponding posterior dimension coordinates. Cell annotations should be used only after fitting: related states should occupy a connected trajectory, and independently chosen marker genes should vary smoothly along the resulting MiSo coordinate. The fourth panel should show the factor co-use graph, with a small number of biologically interpretable hub, branch, and leaf factors.

A companion table should report only the quantities needed to support this picture: held-out deviance relative to Poisson NMF, the average number of active factors per cell, the effective dimensions of the highlighted motifs, stability across random seeds, and external agreement with cell annotations or an independently estimated pseudotime. Thus the intended claim is not that MiSo improves reconstruction at all costs, but that it preserves predictive fit while replacing a high-dimensional loading display by recurring motifs, within-motif coordinates, and a factor-relation graph. The primary display should use fitted motifs before any uncertain merging step; alternative post-processing rules can be reported separately as a sensitivity analysis.

5.5 Ablations and robustness checks

The paper should include ablations for:

1. fixed versus learned F;
2. sensitivity to the number of motifs S;
3. sensitivity to nominal D and the ARD threshold τ;
4. sensitivity to duplicate-factor and correlated-factor strength;
5. stability across random seeds.

These experiments are needed to distinguish the proposed statistical model from an optimization artifact.

6 Discussion

MiSo reframes sparse NMF as a motif discovery problem. Instead of estimating sparse loadings independently for each observation, it pools repeated sparsity patterns through latent motifs. The posterior motif probabilities $\bar{\gamma}_{sdk}$ provide an interpretable uncertainty summary over the factor identities defining each motif, while the responsibilities ω_{is} cluster observations by motif. The empirical-Bayes prior means $\alpha^0_{sd}/\beta^0_{sd}$ provide a natural ARD-style diagnostic for the effective dimension of each motif.

The current evidence supports the promise of this approach, especially in tree/no-anchor simulations and in overfitted-factor uncertainty diagnostics. The main open issues are stable dimension selection, broader multi-seed benchmarks, model-selection guidance for K, S, and D, and a real-data example with scientifically interpretable motifs.

A Threshold-and-recycle initialization

For reproducibility, we record an earlier initialization rule that is retained in the software for historical ablations. Using the motif center c_s and factor ordering r_s defined in the main text, it first forms the ordered set

$$A_s(\tau_{\text{init}}) = (a_{s1}, \ldots, a_{sq_s}) = \{r_s(j) : c_{s,r_s(j)} \geq \tau_{\text{init}}\}.$$

If this set is empty, it is replaced by $A_s = (r_s(1))$. The rule then fills the D dimensions cyclically,

$$\kappa_{sd}^{\text{recycle}} = a_{s,\,1+((d-1) \bmod q_s)}, \qquad d = 1, \ldots, D.$$

When $q_s = 1$, every dimension is assigned the same factor. Conditional on motif s, the resulting factor-space loading is

$$L_i = \sum_{d=1}^{D} \lambda_{id} e_{a_{s1}} = \left(\sum_{d=1}^{D} \lambda_{id} \right) e_{a_{s1}},$$

which lies on a one-dimensional ray regardless of the nominal value of D. More generally, recycling constrains the initialization to at most q_s distinct directions. This can create an artificially low-dimensional starting point when only a few center weights exceed τ_{init}.

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